Abbring & Van den Berg (2003), built up step by step

Plan

1. One duration: survival, hazard, integrated hazard
2. People differ: unobserved heterogeneity and dynamic selection
3. The mixed proportional hazard (MPH) model
4. Two durations racing: competing risks
5. Adding a treatment: potential outcomes indexed by treatment time, no anticipation
6. What we actually observe
7. The model of the paper (Model 1a)
8. Identification, step by step
9. Extensions: heterogeneous effects, multiple spells
10. Connections: DiD with staggered adoption, sequential exogeneity, traps

Running example

- Unemployed people receive benefits.
- The agency can impose a **sanction**: a temporary benefit cut (e.g. for not applying to enough jobs).
- Question: does a sanction make people find a job faster?
- Y: unemployment duration. S: moment the sanction is imposed.

1 One duration

Objects

- $Y \geq 0$: duration (e.g. of unemployment), continuous; y: a value.
- Density $f(y)$, CDF $F(y)$.
- Survival function $S(y) := \Pr(Y > y) = 1 - F(y)$: share still in the state at y.

1.1 Hazard rate

$$\begin{aligned}
\theta(y) &:= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy \mid Y \ge y)}{dy} \\
&= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy)}{dy \ \Pr(Y \ge y)} \\
&= \lim_{dy \downarrow 0} \frac{F(y + dy) - F(y)}{dy} \cdot \frac{1}{S(y)} \\
&= \frac{f(y)}{S(y)} \qquad (1)
\end{aligned}$$

- Line 2: definition of conditional probability.
- Line 3: Y continuous, so $\Pr(Y \ge y) = \Pr(Y > y) = S(y)$.
- Meaning: exit rate *among those still in the state* at y.
- $\theta(y)\,dy \approx$ probability of leaving in $[y, y + dy)$, given still there at y.
- A rate, not a probability: can exceed 1.

1.2 Integrated hazard and survival

Since $S = 1 - F$, we have $S'(y) = -f(y)$. So from (1):

$$\theta(y) = \frac{f(y)}{S(y)} = -\frac{S'(y)}{S(y)} = -\frac{d}{dy} \log S(y).$$

Integrate from 0 to y, using $S(0) = 1$:

$$\begin{aligned}
\Theta(y) := \int_0^y \theta(u)\,du &= -\big[\log S(u)\big]_0^y \\
&= -\log S(y) + \log \underbrace{S(0)}_{=1} \\
&= -\log S(y) \\
\implies \quad S(y) = e^{-\Theta(y)}, &\qquad f(y) = \theta(y)\, e^{-\Theta(y)}. \qquad (2)
\end{aligned}$$

- Θ: integrated (cumulative) hazard.
- θ, Θ, S, f: four descriptions of the same distribution.

1.3 Example: constant hazard

$\theta(y) = c$:

$$\Theta(y) = cy, \qquad S(y) = e^{-cy} \quad \text{(exponential)}$$

$$\Pr(Y > y + u \mid Y > y) = \frac{S(y + u)}{S(y)} = \frac{e^{-c(y+u)}}{e^{-cy}} = e^{-cu}.$$

- Does not depend on y: **memoryless**.
- θ decreasing in y: **negative duration dependence**. Increasing: **positive**.

1.4 Writing Y as a function of a standard random variable

Assume $\theta > 0$, so Θ is strictly increasing and invertible. Define the random variable

$$E := \Theta(Y).$$

Its distribution, for $e \geq 0$:

$$\begin{aligned}
\Pr(E > e) &= \Pr\big(\Theta(Y) > e\big) \\
&= \Pr\big(Y > \Theta^{-1}(e)\big) && (\Theta \text{ strictly increasing}) \\
&= S\big(\Theta^{-1}(e)\big) && (\text{definition of } S) \\
&= \exp\big(-\Theta(\Theta^{-1}(e))\big) && (\text{by (2)}) \\
&= e^{-e}.
\end{aligned}$$

So $E \sim \text{Exp}(1)$ whatever θ is, and inverting the definition of E:

$$Y = \Theta^{-1}(E), \qquad E \sim \text{Exp}(1). \tag{3}$$

Conversely, for any $E \sim \text{Exp}(1)$, $Y := \Theta^{-1}(E)$ has the right distribution:

$$\Pr(Y > y) = \Pr\big(\Theta^{-1}(E) > y\big) = \Pr\big(E > \Theta(y)\big) = e^{-\Theta(y)} = S(y).$$

- Same idea as the probability integral transform: $F(Y) \sim U(0,1)$.
  - Indeed $E = \Theta(Y) = -\log S(Y)$, and $S(Y) = 1 - F(Y) \sim U(0,1)$.
- (3) splits Y into a **model part** Θ and a **standard random part** E.
- Used in Section 5: potential outcomes $Y^*(s) = \Theta_s^{-1}(E)$, where the treatment date s changes Θ only.